iGEM Model / Best Model — how the dry-lab package aligns

iGEM Patras · dry-lab modelling · a self-audit mapping our computational work to the iGEM judging framework, so the wiki and judging-form entries are targeted rather than generic.

What changed in 2026 (why this matters)

Gold is **no longer a separate judge vote**. It is now inferred from the three **Special Awards** a team selects on the Judging Form: each judge scores those awards on their rubrics, and **all three selected awards must reach a "yes"** for the Gold criterion to be met. Teams elect up to three awards — at least one from each list plus a third from either. Selecting fewer than three forfeits Gold eligibility.

Implication for us: if the team selects **Best Model** as one of its three, the modelling work is not decoration — it is load-bearing for the medal. This document is written so that selection is defensible.

> The per-question rubric wording is deliberately unpublished by iGEM (to stop teams "teaching to the test"). This alignment therefore maps our work to the **public** modelling expectations, not to any leaked score language.

The three public modelling expectations, and our evidence

iGEM's public description of strong modelling asks teams to:

(1) Model the system to understand performance and guide experiments.

Our work	How it guides the wet lab
Lipoplex CG-MD (3 formulations × 3 seeds)	Resolves the pH-dependent DNA-binding switch → tells the team to choose an **ionizable** lipid (MC3-type) over permanent-cation DOTAP. A design decision made by the model.
siRNA design + accessibility	Ranks 2,084 candidate sites → 4 orderable duplexes with a defensible priority (CDS1224 first). The model chooses the reagents.
Plasmid audit	

Our work	How it guides the wet lab
	Model-level sequence audit **found a real defect** (plasmid 1 truncated PENK) → removes a construct from the wet-lab plan before money is spent.
Restriction-QC design	Virtual digest predicts the exact gel the wet lab should see → converts modelling into a bench acceptance test.

(2) Show experimental validation in line with project goals.

The model outputs are written as **testable predictions with named assays**, so the wet lab can close the loop: - siRNA → qPCR knockdown at 48 h (predicted ≥70%) + CD10/NEP-activity Western. - Plasmid identity → the two-digest QC gel (predicted band table) + Sanger. - Lipoplex → gel-shift / DLS at endosomal vs physiological pH should mirror the MD contact trend (MC3 binds low-pH, releases neutral; DOTAP binds both).

This is the modelling↔wet-lab **iterative loop** iGEM rewards — each model output names the experiment that would confirm or refute it.

(3) Consider future implementation — application, scalability, safety.

- **Application:** intrathecal lipoplex delivery to the spinal dorsal horn for analgesia (stated, with the BBB rationale).
- **Scalability:** the pipeline is a reusable code repository (below), not a one-off — any PENK/NEP construct can be re-audited.
- **Safety:** limitations are stated by design, not hidden — NEP off-target substrates (Aβ, substance P) motivate keeping knockdown local/ transient; MD models are MARTINI analogies so conclusions are comparative, not affinities.

Why this modelling is credible (the differentiators)

Judges reward modelling that is honest and reproducible over modelling that overclaims. Our package is built to that standard:

1. **A real negative result.** The audit found and reported a genuine defect (plasmid 1) rather than rubber-stamping all constructs. Finding problems is evidence the audit is real.
2. **Error bars and convergence honesty.** MD metrics are reported as mean ± SEM over 3 seeds, and the large error bars are acknowledged (binding still nucleating at 50 ns) — which is why the 200 ns extension is running. We report uncertainty instead of a single pretty number.
3. **Stated model limitations.** MARTINI topologies are analogy models; accessibility folding ignores RNA-binding proteins; seed-level off-target frequency is flagged as a follow-up. Each caveat is written next to the result it qualifies.

4. **Full reproducibility.** Every method, force field, database, and design rule is traceable to a methods/references log, and the entire codebase is in a version-controlled repository (pushed to GitHub).

Deliverables checklist for the Best Model wiki page

To convert this into a scoring wiki page and judging-form entry:

- [ ] **Model page** narrating the four workstreams as models that made decisions (lead with the plasmid-1 defect and the MC3-vs-DOTAP choice).
- [ ] Embed the figures: `restriction_qc_gel.png`, `sirna_design_v2_figure.png`, `lipoplex_md_50ns.png` (→ 200 ns when it lands).
- [ ] A **methods & reproducibility** section linking the GitHub repo and the methods log — reproducibility is a scored differentiator.
- [ ] The **model→experiment loop** table (expectation 2 above) shown explicitly, so judges see validation is planned, not hoped for.
- [ ] A **limitations** box — counterintuitively a positive signal to judges.
- [ ] 150-word judging-form description: lead with "our models made three concrete design decisions the wet lab acted on."

One-paragraph pitch (drop-in for the 150-word form)

> Our dry-lab did not describe the system — it decided it. A sequence-level audit of four PENK cassettes found a real defect (a truncated enkephalin core) and selected the construct to build. A ViennaRNA-scored siRNA pipeline screened 2,084 sites to four orderable anti-neprilysin duplexes with a ranked priority. Coarse-grained MARTINI MD across three lipid formulations (triplicate seeds, mean ± SEM) resolved the pH-dependent DNA-binding switch that motivates our ionizable-lipid choice over a permanent cation. Every model output is written as a testable prediction with a named wet-lab assay, closing the design–build–test loop. Limitations — MARTINI analogy topologies, in-cell RNA structure, off-target seed frequency — are stated beside each result. The full pipeline is reproducible and version-controlled. The model is load-bearing: remove it and three wet-lab decisions lose their basis.